

PROGRAM A: SEQUENTIAL FILE ORGANIZATION

C PROGRAM:

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main() {
    FILE *fp;
    struct student s;
    fp = fopen("student.dat", "w");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fprintf(fp, "%d %s\n", s.regno, s.name);
    fclose(fp);
    fp = fopen("student.dat", "r");
    fscanf(fp, "%d %s", &s.regno, s.name);
    printf("\nRecord Details\n");
    printf("Register Number : %d\n", s.regno);
    printf("Name : %s\n", s.name);
    fclose(fp);
    return 0;
}
```

SHELL SCRIPT:

```
#!/bin/bash

echo "Enter Register Number:"

read regno

echo "Enter Name:"

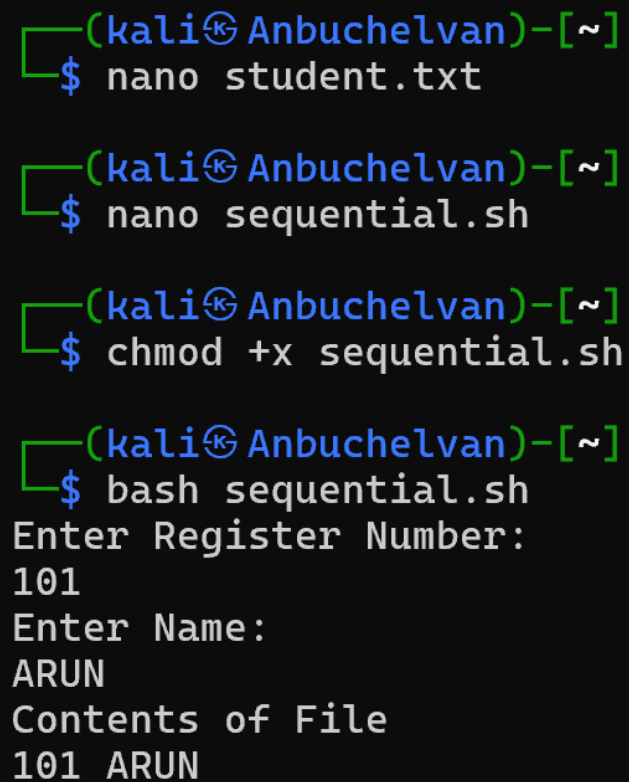
read name

echo "$regno $name" > student.txt

echo "Contents of File"

cat student.txt
```

OUTPUT:



```
(kali㉿ Anbuchelvan)-[~]
$ nano student.txt

(kali㉿ Anbuchelvan)-[~]
$ nano sequential.sh

(kali㉿ Anbuchelvan)-[~]
$ chmod +x sequential.sh

(kali㉿ Anbuchelvan)-[~]
$ bash sequential.sh
Enter Register Number:
101
Enter Name:
ARUN
Contents of File
101 ARUN
```

B: DIRECT (RANDOM) FILE ORGANIZATION

C PROGRAM:

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    FILE *fp;
    struct student s;
    fp = fopen("random.dat", "wb+");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fwrite(&s, sizeof(s), 1, fp);
    rewind(fp);
    fread(&s, sizeof(s), 1, fp);
    printf("\nRecord Found\n");
    printf("Reg No : %d\n", s.regno);
    printf("Name : %s\n", s.name);
    fclose(fp);
    return 0;
}
```

SHELL SCRIPT:

```
#!/bin/bash
```

```
echo "Enter Record:"  
  
read rec  
  
echo "$rec" > random.txt  
  
echo "Random Access Record"  
  
sed -n '1p' random.txt
```

OUTPUT:

```
(kali㉿ Anbuchelvan)-[~]  
$ nano random.txt
```

```
(kali㉿ Anbuchelvan)-[~]  
$ nano random.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ chmod +x random.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ bash random.sh
```

Enter Record:

12

Random Access Record

12

C: INDEXED FILE ORGANIZATION

C PROGRAM:

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

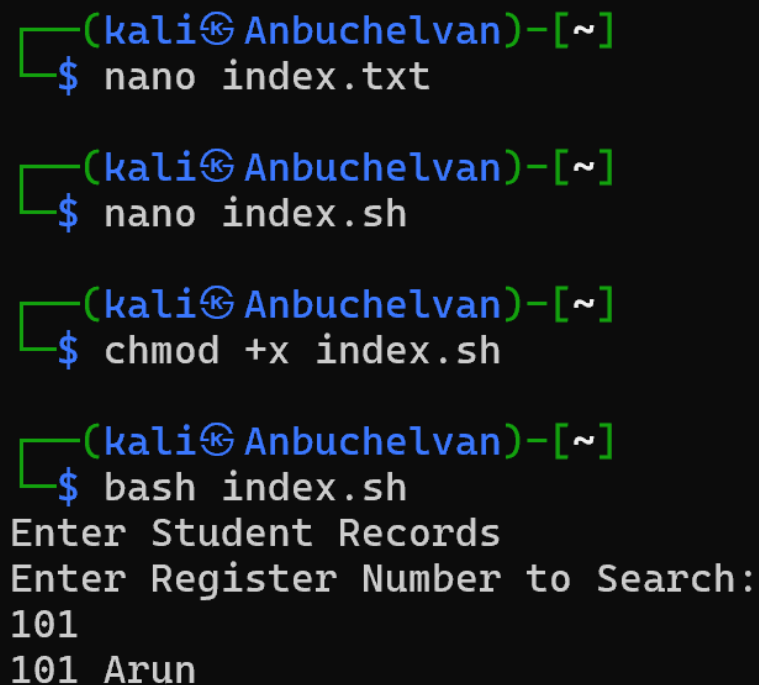
int main()
{
    struct student s[3];
    int key, i;
    printf("Enter 3 Student Records\n");
    for(i=0;i<3;i++)
    {
        scanf("%d %s",&s[i].regno,s[i].name);
    }
    printf("Enter Register Number to Search: ");
    scanf("%d",&key);
    for(i=0;i<3;i++)
    {
        if(s[i].regno==key)
        {
            printf("\nRecord Found\n");
            printf("Reg No : %d\n",s[i].regno);
            printf("Name : %s\n",s[i].name);
        }
    }
    return 0;
}
```

```
}  
  
printf("Record Not Found\n");  
  
return 0;  
  
}
```

SHELL SCRIPT:

```
#!/bin/bash  
  
echo "Enter Student Records"  
  
echo "101 Arun" > index.txt  
  
echo "102 Kumar" >> index.txt  
  
echo "103 Ravi" >> index.txt  
  
echo "Enter Register Number to Search:"  
  
read key  
  
grep "^$key" index.txt
```

OUTPUT:



```
(kaliⓈ Anbuchelvan)-[~]  
$ nano index.txt  
  
(kaliⓈ Anbuchelvan)-[~]  
$ nano index.sh  
  
(kaliⓈ Anbuchelvan)-[~]  
$ chmod +x index.sh  
  
(kaliⓈ Anbuchelvan)-[~]  
$ bash index.sh  
Enter Student Records  
Enter Register Number to Search:  
101  
101 Arun
```